공학수학(2)

11.6 Cylinders and Quadric Surfaces

Cylinders and Quadric Surfaces

We have already looked at two special types of surfaces: planes and spheres.

Here we investigate two other types of surfaces: cylinders and quadric surfaces.

In order to sketch the graph of a surface, it is useful to determine the curves of intersection of the surface with planes parallel to the coordinate planes.

These curves are called **traces** (or cross-sections) of the surface.

Cylinders

A **cylinder** is a surface that consists of all lines (called **rulings**) that are parallel to a given line and pass through a given plane curve.

Example 1

Sketch the graph of the surface $z = x^2$.

Solution:

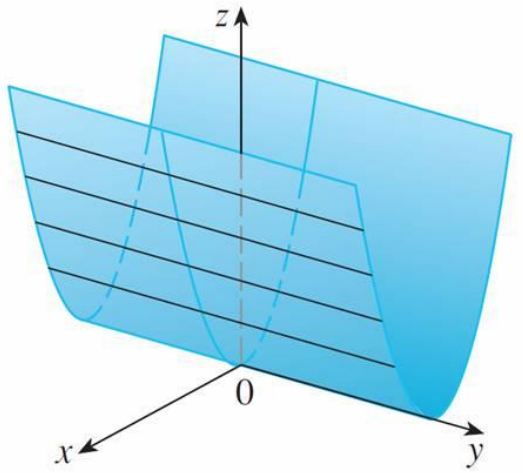
Notice that the equation of the graph, $z = x^2$, doesn't involve y.
This means that any vertical plane with equation $y = k$(parallel to the xz-plane) intersects the graph in a curve with equation $z = x^2$.

So these vertical traces are parabolas.

Example 1 – Solution

Figure 1 shows how the graph is formed by taking the parabola $z = x^2$ in the *xz*-plane and moving it in the direction of the *y*-axis.

The graph is a surface, called a **parabolic cylinder**, made up of infinitely many shifted copies of the same parabola. Here the rulings of the cylinder are parallel to the *y*-axis.



The surface $z = x^2$ 쟤 a parabolic cylinder

Figure 1

Cylinders

In Example 1 the variable *y* is missing from the equation of the cylinder. This is typical of a surface whose rulings are parallel to one of the coordinate axes. If one of the variables *x*, *y* or *z* is missing from the equation of a surface, then the surface is a cylinder.

Note

When you are dealing with surfaces, it is important to recognize that an equation like $x^2 + y^2 = 1$ represents a cylinder and not a circle. The trace of the cylinder $x^2 + y^2 = 1$ in the *xy*-plane is the circle with equations $x^2 + y^2 = 1,\ z = 0$.

Quadric Surfaces

A **quadric surface** is the graph of a second-degree equation in three variables x, y, and z. The most general such equation is

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + Gx + Hy + Iz + J = 0$$

where $A,B,C,\ldots, J$ are constants, but by translation and rotation it can be brought into one of the two *standard forms*

$$Ax^2 + By^2 + Cz^2 + J = 0 \quad \text{or} \quad Ax^2 + By^2 + Iz = 0$$

Quadric surfaces are the counterparts in three dimensions of the conic sections in the plane.

Example 3

Use traces to sketch the quadric surface with equation

$$x^2 + \frac{y^2}{9} + \frac{z^2}{4} = 1$$

Solution:

By substituting $z = 0$, we find that the trace in the xy-plane is $x^2 + y^2/9 = 1$, which we recognize as an equation of an ellipse. In general, the horizontal trace in the plane $z = k$ is

$$x^2 + \frac{y^2}{9} = 1 - \frac{k^2}{4} \qquad z = k$$

which is an ellipse, provided that $k^2 < 4$, that is, $-2 < k < 2$.

Example 3 – Solution

(If $|k| = 2,$ the trace consists of a single point, and the trace is empty for $|k| > 2.$)

Similarly, vertical traces parallel to the *yz*- and *xz*-planes are also ellipses:

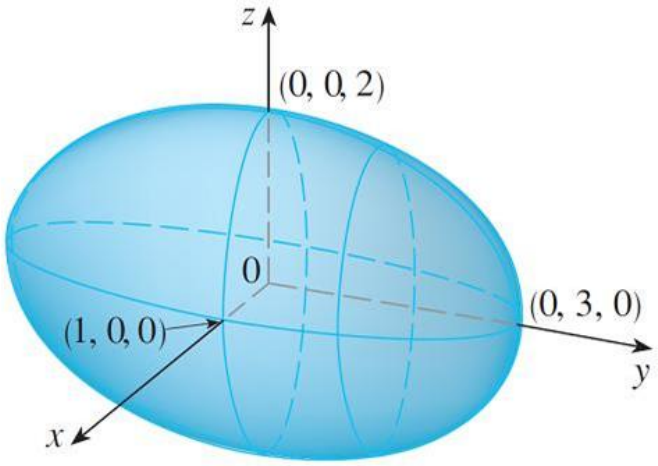
$$\frac{y^2}{9} + \frac{z^2}{4} = 1 - k^2 \qquad x = k \qquad (\text{if } -1 < k < 1)$$

$$x^2 + \frac{z^2}{4} = 1 - \frac{k^2}{9} \qquad y = k \qquad (\text{if } -3 < k < 3)$$

Example 3 – Solution

Figure 4 shows how drawing some traces indicates the shape of the surface. It's called an **ellipsoid** because all of its traces are ellipses.

Notice that it is symmetric with respect to each coordinate plane; this is because its equation involves only even powers of x, y, and z.



The ellipsoid $x^2 + \frac{y^2}{9} + \frac{z^2}{4} = 1$

Figure 4

Example 4

Use traces to sketch the surface $z = 4x^2 + y^2$.

Solution:

If we put $x = 0$, we get $z = y^2$, so the yz-plane intersects the surface in a parabola. If we put $x = k$ (a constant), we get $z = y^2 + 4k^2$.

This means that if we slice the graph with any plane parallel to the yz-plane, we obtain a parabola that opens upward.

Similarly, if $y = k$, the trace is $z = 4x^2 + k^2$, which is again a parabola that opens upward.

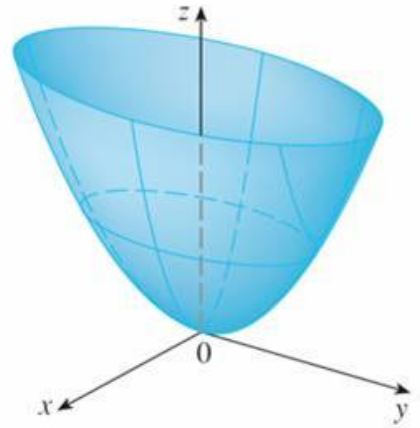
Example 4 – Solution

If we put $z = k$, we get the horizontal traces $4x^2 + y^2 = k$, which we recognize as a family of ellipses ($k > 0$). Knowing the shapes of the traces, we can sketch the graph in Figure 5.

Because of the elliptical and parabolic traces, the quadric surface

$$z = 4x^2 + y^2,$$

is called an **elliptic paraboloid**.



The surface $z = 4x^2 + y^2$ is an elliptic paraboloid.
Horizontal traces are ellipses; vertical traces are parabolas.

Figure 5

Example 5

Sketch the surface $z = y^2 - x^2$.

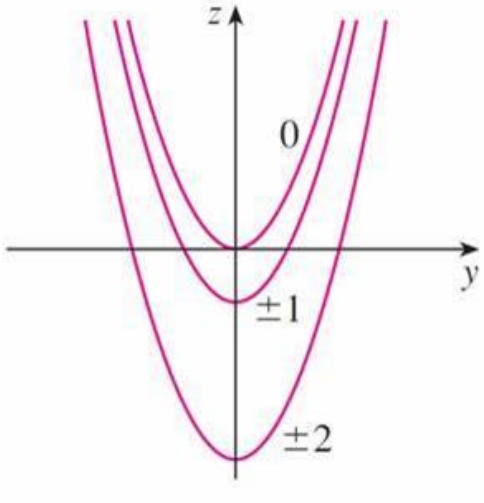
Solution:

The traces in the vertical planes $x = k$ are the parabolas $z = y^2 - k^2$, which open upward. The traces in $y = k$ are the parabolas $z = -x^2 + k^2$, which open downward.

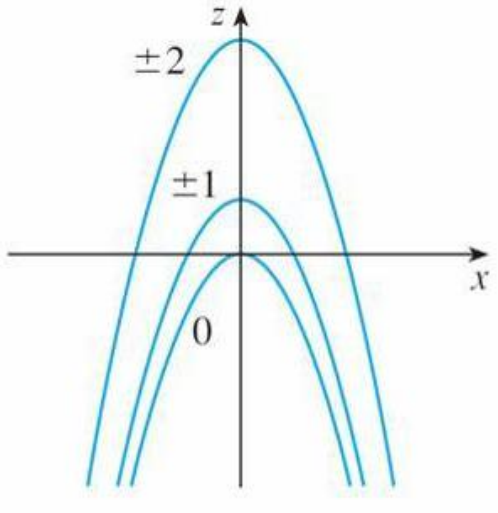
The horizontal traces are $y^2 - x^2 = k$, a family of hyperbolas.

Example 5 – Solution

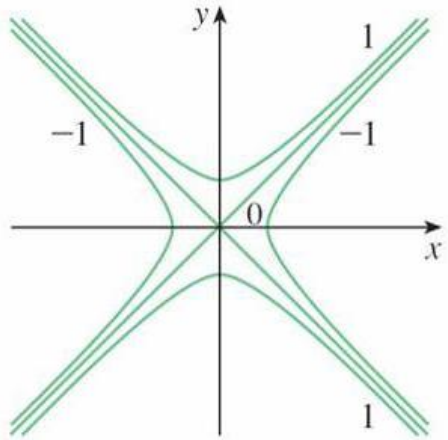
We draw the families of traces in Figure 6, and we show how the traces appear when placed in their correct planes in Figure 7.



Traces in $x = k$ are $z = y^2 - k^2$.



Traces in $y = k$ are $z = -x^2 + k^2$.

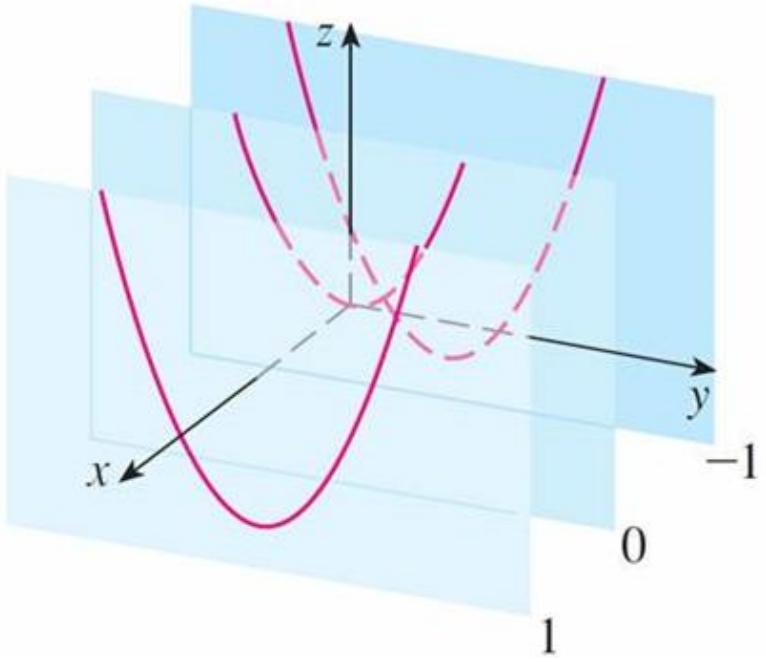


Trace in $z = k$ are $y^2 - x^2 = k$.

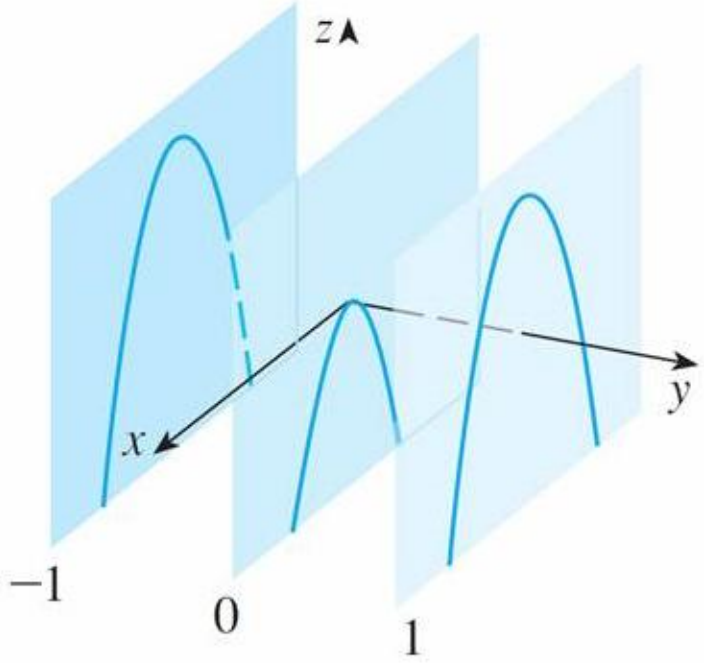
Vertical traces are parabolas; horizontal traces are hyperbolas. All traces are labeled with the value of k.

Figure 6

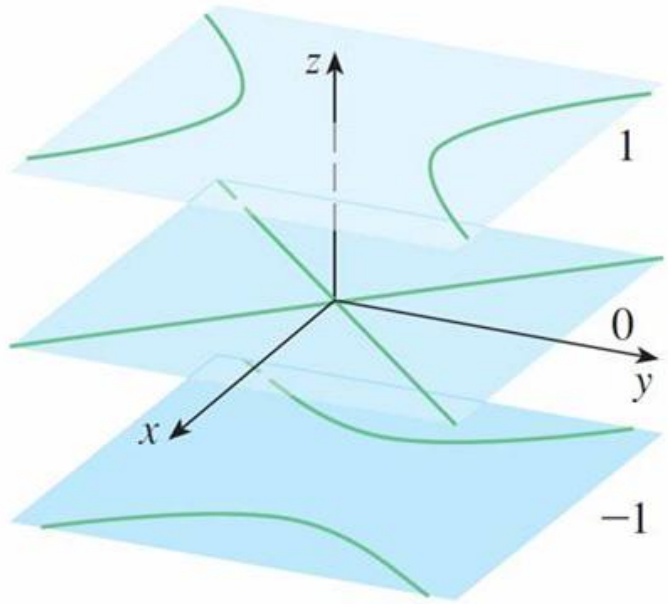
Example 5 – Solution



Traces in $x = k$



Traces in $y = k$



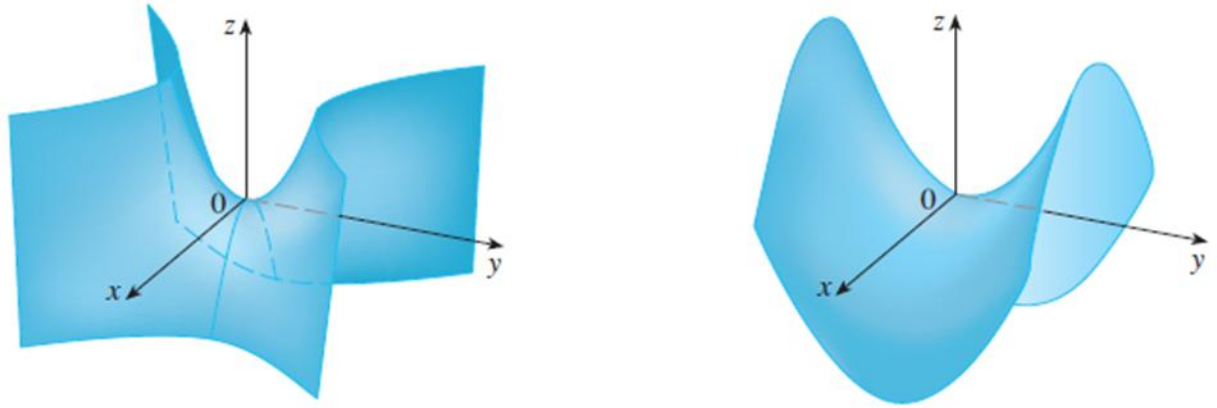
Traces in $z = k$

Traces moved to their correct planes

Figure 7

Example 5 – Solution

In Figure 8 we fit together the traces from Figure 7 to form the surface $z = y^2 - x^2$, a **hyperbolic paraboloid**.



Two views of the surface $z = y^2 - x^2$, a hyperbolic paraboloid.

Figure 8

Notice that the shape of the surface near the origin resembles that of a saddle.

Example 6

Sketch the surface $\frac{x^2}{4} + y^2 - \frac{z^2}{4} = 1$.

Solution:

The trace in any horizontal plane $z = k$ is the ellipse

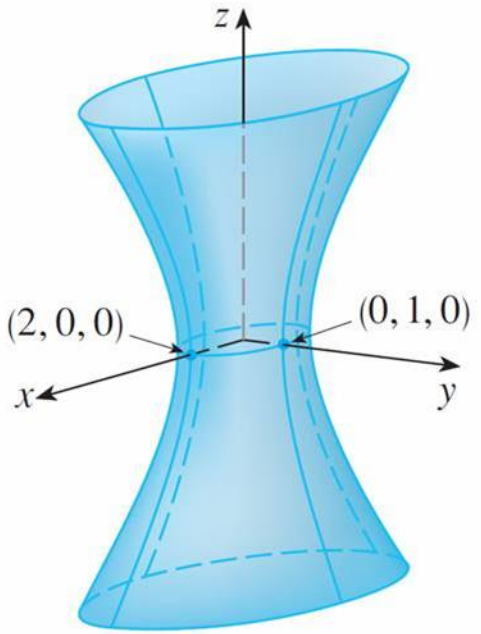
$$\frac{x^2}{4} + y^2 = 1 + \frac{k^2}{4} \qquad z = k$$

but the traces in the *xz*- and *yz*-planes are the hyperbolas

$$\frac{x^2}{4} - \frac{z^2}{4} = 1 \quad y = 0 \quad \text{and} \quad y^2 - \frac{z^2}{4} = 1 \quad x = 0$$

Example 6 – Solution

This surface is called a **hyperboloid of one sheet** and is sketched in Figure 9.



The surface $\frac{x^2}{4} + y^2 - \frac{z^2}{4} = 1,$

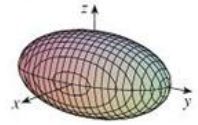
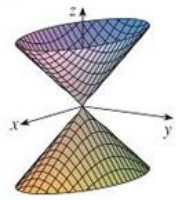
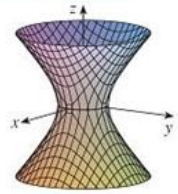
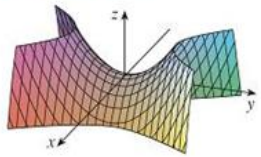
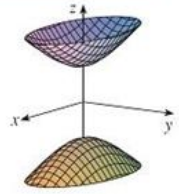
a hyperboloid of one sheet

Figure 9

Quadric Surfaces

Table 1 shows computer-drawn graphs of the six basic types of quadric surfaces in standard form.

All surfaces are symmetric with respect to the *z*-axis. If a quadric surface is symmetric about a different axis, its equation changes accordingly.

Surface	Equation	Surface	Equation
Ellipsoid 	$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ All traces are ellipses. If $a=b=c$, the ellipsoid is a sphere.	Cone 	$\frac{z^2}{c^2}=\frac{x^2}{a^2}+\frac{y^2}{b^2}$ Horizontal traces are ellipses. Vertical traces in the planes $x=k$ and $y=k$ are hyperbolas if $k\neq 0$ but are pairs of lines if $k=0$.
Elliptic Paraboloid 	$\frac{z}{c}=\frac{x^2}{a^2}+\frac{y^2}{b^2}$ Horizontal traces are ellipses. Vertical traces are parabolas. The variable raised to the first power indicates the axis of the paraboloid.	Hyperboloid of One Sheet 	$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$ Horizontal traces are ellipses. Vertical traces are hyperbolas. The axis of symmetry corresponds to the variable whose coefficient is negative.
Hyperbolic Paraboloid 	$\frac{z}{c}=\frac{x^2}{a^2}-\frac{y^2}{b^2}$ Horizontal traces are hyperbolas. Vertical traces are parabolas. The case where $c<0$ is illustrated.	Hyperboloid of Two Sheets 	$-\frac{x^2}{a^2}-\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ Horizontal traces in $z=k$ are ellipses if $k>c$ or $k<-c$. Vertical traces are hyperbolas. The two minus signs indicate two sheets.

Graphs of Quadric Surfaces

Table 1